

# Notes: Gatev, Schuermann, and Strahan (RFS, 2009)

## Managing Bank Liquidity Risk: How Deposit-Loan Synergies Vary with Market Conditions

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# 1 Big picture

- **Question.** Can transaction deposits reduce banks' liquidity risk from unused loan commitments, and does this deposit-loan hedge become stronger when market liquidity is tight?
- **Core puzzle.** Banks provide liquidity on both sides of the balance sheet. Depositors can withdraw funds, while borrowers can draw on unused loan commitments. Traditional bank-run theories emphasize transactions deposits as a source of fragility. This paper asks whether transactions deposits can instead be a stabilizing liquidity hedge.
- **Main theory background.** The paper tests the key implication of Kashyap, Rajan, and Stein (2002): because liquidity demand from depositors and borrowers is imperfectly correlated, combining deposits and commitments can reduce banks' need to hold cash and can lower liquidity risk.
- **Empirical setting.** The sample contains large publicly traded U.S. banks from 1990 to 2002. The outcome is bank stock-return volatility, estimated weekly from daily stock returns using GARCH or realized volatility measures.
- **Main empirical result.** Stock-return volatility increases with unused loan commitments for banks with few transaction deposits, but this relationship is much weaker or absent for banks with many transaction deposits.
- **Key interaction.** The coefficient on the interaction between unused commitments and transaction deposits is negative and significant across many specifications. This is the main evidence of a deposit-loan liquidity hedge.
- **Market-condition result.** The hedge becomes stronger when market liquidity is tight, especially when the commercial-paper minus Treasury-bill spread is high and during the October 1998 liquidity crisis.
- **Interpretation.** In tight markets, borrowers draw on commitments while nervous investors move funds into bank deposits. These opposite liquidity flows make transactions deposits especially valuable as a hedge against asset-side liquidity demand.
- **Contribution to bank-risk thinking.** The paper reverses the usual intuition that transactions deposits primarily create liquidity risk. Deposits can also be a risk-management asset for banks that offer loan commitments.
- **Takeaway.** Liquidity production on both sides of the balance sheet is not merely a source of fragility. It can create a diversification synergy that is most powerful when outside liquidity is scarce.

## 2 Incremental contribution of this paper

### 2.1 Contribution relative to classic bank-run theory

- Classic Diamond-Dybvig-style thinking emphasizes demand deposits as fragile liabilities that can trigger runs.
- Gatev, Schuermann, and Strahan show that the same transaction deposits can stabilize banks by providing an inflow of funds when market-wide liquidity becomes scarce.
- The paper therefore adds a second view of deposits: deposits are not only runnable liabilities; they can be a natural hedge for liquidity commitments on the asset side.

**Interpretation:** The paper changes the interpretation of deposit liquidity risk. It does not deny that deposits can be fragile, but it shows that deposits also help banks manage liquidity risk from borrower drawdowns.

## 2.2 Contribution relative to Kashyap, Rajan, and Stein (2002)

- KRS develop the theoretical idea that deposits and loan commitments should be jointly produced because their liquidity demands are not perfectly correlated.
- KRS show that banks with more unused commitments also tend to have more transactions deposits.
- This paper tests the key missing implication: whether combining commitments and deposits actually reduces bank risk.
- The authors use stock-return volatility as a market-based measure of bank risk and show that transaction deposits attenuate the volatility effect of commitments.

**Interpretation:** The incremental step is moving from a balance-sheet correlation to a direct risk-reduction test. The paper asks whether the predicted hedge is visible in market prices.

## 2.3 Contribution relative to liquidity-crisis research

- Many papers study how market liquidity shocks affect loan supply or bank balance sheets.
- This paper shows that the deposit-loan hedge is state-dependent: it is much stronger when commercial-paper spreads are high.
- The 1998 crisis provides a sharp setting in which market borrowers draw on back-up lines while investors move cash into banks.

**Interpretation:** The paper links bank specialness to crisis states. Banks are especially valuable when market liquidity dries up because their deposit base absorbs liquidity flight.

## 2.4 Contribution relative to empirical bank-risk measurement

- The paper uses high-frequency stock-return volatility instead of accounting risk or failure outcomes.
- It estimates conditional volatility using bank-year GARCH models, then validates the results with realized volatility and residual volatility.
- It controls for market volatility, short rates, commercial-paper spreads, liquid assets, size, fed-funds borrowing, equity capital, credit risk, and market-risk measures.

**Interpretation:** The empirical design makes the liquidity-hedge result hard to attribute to ordinary market risk, interest-rate risk, credit risk, or bank size.

# 3 Data and measurement

## 3.1 Bank sample

- The sample covers large publicly traded U.S. banks from 1990 to 2002.
- The authors begin in 1990 because unused retail loan commitments are available from that year onward.
- Each year, the sample is based on the largest banks with publicly traded equity and valid stock-return information.
- Banks involved in mergers or acquisitions are dropped during the deal year to avoid volatility generated by merger speculation.
- The resulting sample has 170 banks and roughly 49,500 bank-week observations.

**Interpretation:** Large traded banks are used because their stocks trade frequently, their returns can be used to estimate volatility, and they hold most banking-system assets and loan commitments.

### 3.2 Loan commitment exposure

The main asset-side liquidity exposure measure is:

$$LoanCommitments_{i,t} = \frac{Unused\ commitments_{i,t}}{Unused\ commitments_{i,t} + Loans_{i,t}}. \quad (1)$$

The paper also uses a version that excludes retail commitments:

$$BusinessCommitments_{i,t} = \frac{Commitments_{i,t} - RetailCommitments_{i,t}}{Commitments_{i,t} - RetailCommitments_{i,t} + Loans_{i,t}}. \quad (2)$$

**Interpretation:** Unused loan commitments expose banks to liquidity risk because firms can draw down credit lines on demand, especially when market funding becomes difficult.

### 3.3 Transaction deposit base

The main liability-side liquidity measure is:

$$DepositBase_{i,t} = \frac{Transactions\ deposits_{i,t}}{Total\ deposits_{i,t}}. \quad (3)$$

- Transaction deposits include deposits that households and firms use for payments and liquidity management.
- These deposits can flow into banks when investors seek safety or liquidity.
- The measure is taken from quarterly Call Reports and lagged relative to the weekly volatility outcome.

**Interpretation:** The paper’s central hypothesis is that this deposit base hedges unused commitment drawdowns.

### 3.4 Stock-return volatility

The main dependent variable is bank stock-return volatility. The first-stage GARCH model is:

$$r_{i,t+1} = \mu_i + \varepsilon_{i,t+1}, \quad (4)$$

$$\sigma_{i,t+1}^2 = \omega + \alpha \varepsilon_{i,t}^2 + \beta \sigma_{i,t}^2, \quad \omega, \alpha, \beta > 0. \quad (5)$$

Daily conditional volatilities are aggregated to weekly frequency and annualized. Robustness tests replace the GARCH volatility with realized volatility and realized residual volatility.

**Interpretation:** Stock volatility is a market-based risk measure. If commitments raise liquidity risk, equity volatility should rise. If deposits hedge that risk, the commitment-volatility relation should be weaker for deposit-rich banks.

### 3.5 Market and bank controls

The regressions include market-level controls:

- GARCH volatility of the S&P 500;
- the commercial-paper minus Treasury-bill spread;
- the 3-month Treasury-bill yield.

They also include bank-level controls:

- log assets;
- cash plus securities divided by assets;
- Fed funds purchased divided by assets;
- equity divided by assets;
- credit and market risk controls in robustness tests.

**Interpretation:** These controls isolate the deposit-loan liquidity hedge from general market volatility, interest rates, credit spreads, size, balance-sheet liquidity, and capital adequacy.

## 4 Empirical specifications

### 4.1 Main regression

The baseline regression is:

$$\log(\sigma_{i,t}) = \beta_0 + \beta_1 \text{LoanCommitments}_{i,t-1} + \beta_2 \text{DepositBase}_{i,t-1} \quad (6)$$

$$+ \beta_3 (\text{LoanCommitments}_{i,t-1} \times \text{DepositBase}_{i,t-1}) + \Gamma' X_{i,t-1} + \Lambda' M_t + u_{i,t}. \quad (7)$$

- $\sigma_{i,t}$  is weekly conditional stock-return volatility.
- $X_{i,t-1}$  contains lagged bank characteristics.
- $M_t$  contains market-level volatility and interest-rate controls.
- Standard errors are clustered by bank.

**Interpretation:** The key coefficient is  $\beta_3$ . The deposit-loan hedge hypothesis predicts  $\beta_3 < 0$ .

### 4.2 Alternative volatility models

The paper checks that results are not driven by the GARCH model by using:

$$\text{RealizedVolatility}_{i,t} = \log \left( \frac{1}{D_t} \sum_{d \in t} r_{i,d}^2 \right), \quad (8)$$

where  $D_t$  is the number of trading days in the week.

It also estimates a factor-adjusted residual volatility model:

$$r_{i,t} = \alpha_i + \beta_{1,i} r_{m,t} + \beta_{2,i} \Delta(\text{Baa} - \text{Aaa})_t + \beta_{3,i} \Delta(\text{TBill})_t + \varepsilon_{i,t}, \quad (9)$$

then uses the weekly average squared residuals.

**Interpretation:** These tests show that the result is not simply caused by market beta, credit risk, or interest-rate exposure.

### 4.3 Between and within variation

The paper decomposes the evidence into:

- between-bank variation, using bank-level time-series averages;
- within-bank variation, using deviations from bank-level means.

**Interpretation:** The hedge appears in both cross-sectional bank differences and time-series changes, reducing the concern that it is driven only by persistent unobserved bank traits.

#### 4.4 Market-condition tests

The paper estimates regime-specific regressions for:

- low commercial-paper spread weeks;
- high commercial-paper spread weeks;
- the first three weeks of October 1998, when spreads rose above 100 basis points.

It also estimates regressions with interactions between loan commitments, transaction deposits, and the paper-bill spread:

$$\log(\sigma_{i,t}) = \dots + \delta_1 \text{LoanCommitments}_{i,t-1} \times \text{Spread}_t + \delta_2 \text{DepositBase}_{i,t-1} \times \text{Spread}_t + \delta_3 \text{LoanCommitments}_{i,t-1} \times \text{DepositBase}_{i,t-1} \times \text{Spread}_t \quad (10)$$

**Interpretation:** If the result is truly a liquidity hedge, it should become stronger when market liquidity is scarce. Tables 8 and 9 show exactly this.

### 5 Identification and interpretation

#### 5.1 Main identification logic

- The paper uses lagged balance-sheet exposure to explain weekly stock-return volatility.
- The interaction term identifies whether transaction deposits attenuate the volatility effect of unused commitments.
- The market-condition tests exploit high-frequency changes in outside liquidity conditions, while balance-sheet variables move only quarterly.

**Interpretation:** The state-dependence of the hedge helps address reverse causality. Banks do not rapidly choose their deposit-commitment mix week by week, but the hedge becomes stronger when market liquidity tightens.

#### 5.2 Reverse-causality concern

A concern is that safer banks choose both more deposits and more commitments. The paper addresses this by:

- controlling for size, liquid assets, fed-funds borrowing, equity capital, credit risk, and market risk;
- showing the interaction survives within-bank variation;
- showing the hedge strengthens in high-spread weeks and during October 1998.

**Interpretation:** The tight-market evidence is the strongest response: the same balance-sheet mix provides more hedging exactly when theory predicts liquidity demands become negatively correlated.

#### 5.3 Economic interpretation

- Loan commitments create asset-side liquidity exposure because borrowers draw funds when external markets become illiquid.

- Transaction deposits create liability-side liquidity inflows when households and firms move funds into banks for safety and payments.
- When the two flows occur in opposite states, banks can economize on cash and reduce risk.

**Interpretation:** Banks are liquidity insurers to both borrowers and depositors. The synergy comes from serving both sides jointly.

## 6 Tables and figures

This section keeps adjacent tables and figures on the same row wherever possible. If a visual appears alone, it is shown at width 0.6.

### 6.1 Table 1 and Figure 1: basic deposit-loan hedge

**Read:**

- Table 1 sorts banks by unused commitments and transactions deposits.
- Among banks with low transaction deposits, volatility rises from 0.28 to 0.36 as commitments rise.
- Among banks with high transaction deposits, volatility does not rise with commitments; it is about 0.31–0.32.
- Figure 1 shows the same pattern graphically: a positive commitment-volatility slope for low-deposit banks but a flat/negative slope for high-deposit banks.

**Economic message:** Transactions deposits hedge liquidity risk from loan commitments. Commitment exposure is risky mainly for banks without a strong deposit base.

**Table 1**  
Mean bank characteristics, by transactions deposits and unused commitments

|                                                      | Transactions deposits/total deposits (TD) |                                 |                     |
|------------------------------------------------------|-------------------------------------------|---------------------------------|---------------------|
|                                                      | TD < 33rd percentile                      | 33rd pct. $\geq$ TD < 67th pct. | 67th pct. $\leq$ TD |
|                                                      | (1)                                       | (2)                             | (3)                 |
| <i>Unused commitments/(commitments + loans) (LC)</i> |                                           |                                 |                     |
| LC < 33rd percentile                                 |                                           |                                 |                     |
| Stock-return volatility                              | 0.28                                      | 0.29                            | 0.32                |
| (Cash + securities)/assets                           | 0.32                                      | 0.31                            | 0.33                |
| Assets (billions of \$)                              | 10.58                                     | 10.20                           | 7.14                |
| Fed funds purchased/assets                           | 0.07                                      | 0.08                            | 0.07                |
| Equity/assets                                        | 0.08                                      | 0.08                            | 0.11                |
| 33rd percentile $\leq$ LC < 67th percentile          |                                           |                                 |                     |
| Stock-return volatility                              | 0.29                                      | 0.29                            | 0.30                |
| (Cash + securities)/assets                           | 0.32                                      | 0.31                            | 0.33                |
| Assets (billions of \$)                              | 17.85                                     | 21.64                           | 16.53               |
| Fed funds purchased/assets                           | 0.09                                      | 0.10                            | 0.11                |
| Equity/assets                                        | 0.08                                      | 0.08                            | 0.07                |
| 67th percentile $\leq$ LC                            |                                           |                                 |                     |
| Stock-return volatility                              | 0.36                                      | 0.32                            | 0.31                |
| (Cash + securities)/assets                           | 0.34                                      | 0.30                            | 0.33                |
| Assets (billions of \$)                              | 34.63                                     | 89.10                           | 83.36               |
| Fed funds purchased/assets                           | 0.07                                      | 0.10                            | 0.10                |
| Equity/assets                                        | 0.08                                      | 0.08                            | 0.08                |
| Mean commitments ratio                               | 0.30                                      | 0.31                            | 0.37                |

Average bank characteristics and stock-return volatility, equal to the annualized return standard deviation, for large, publicly traded banks. Data on unused commitments, transactions deposits, as well as the other bank characteristics, come from the most recent quarter of the Reports of Income and Condition *prior* to the time at which we measure the stock-return variability. The observations are sorted vertically by the percentiles of the distribution of level of loan commitments (unused commitments divided by unused commitments plus loans) and horizontally by the level of transactions deposits relative to total deposits. Each of the nine cells reports

Table 1: bank characteristics by deposits and commitments

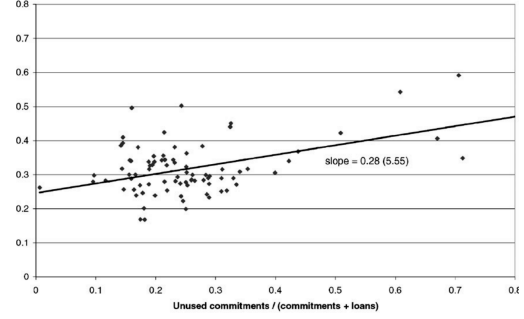


Figure 1a  
Stock Return Volatility for Low Transactions Deposit Banks, 1990-2002

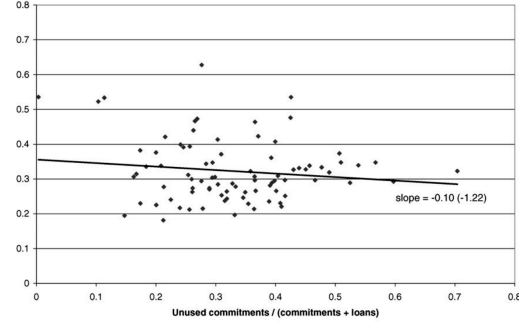


Figure 1b  
Stock Return Volatility for High Transactions Deposit Banks, 1990-2002

Figure 1: commitment exposure and stock-return volatility

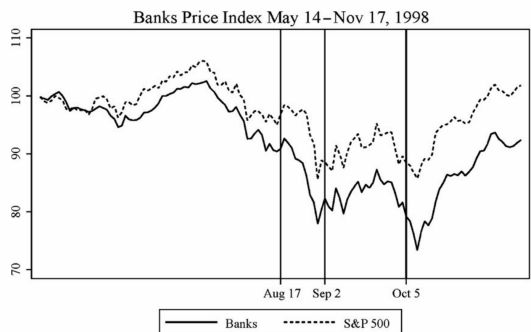
## 6.2 Figure 2 and Table 2: 1998 liquidity crisis and summary statistics

Read:

- Figure 2 shows that bank stock prices fell sharply and bank volatility rose during the 1998 crisis.
- Bank conditional volatility rose along with market stress, especially around late August and early October.
- Table 2 reports the mean bank volatility of 0.30, mean commitment ratio of 0.32, and mean transaction-deposit ratio of 0.26.

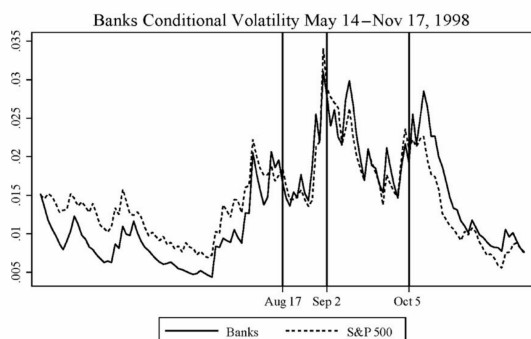
**Economic message:** The paper studies a setting where liquidity stress is visible in bank equity markets and where banks have substantial exposure to both commitments and transaction deposits.





Daily index data from CRSP. The bank index is equal-weighted. The index levels are normalized to 100 on May 14, 1998.

**Figure 2a**  
Bank Stock-Price Index.



Daily data from CRSP. Conditional volatilities from the EGARCH (1,1) model.

**Figure 2b**  
Bank Conditional Stock-Return Volatility.

*Figure 2: bank stock index and conditional volatility in 1998*

**Table 2**  
Summary statistics for variables included in regression models

|                                                                                   | Mean  | Standard deviation |
|-----------------------------------------------------------------------------------|-------|--------------------|
|                                                                                   | (1)   | (2)                |
| <i>Dependent variable</i>                                                         |       |                    |
| Stock-return volatility<br>(annualized return standard deviation)                 | 0.30  | 0.13               |
| <i>Commitments and deposits</i>                                                   |       |                    |
| Commitments/(commitments + loans)                                                 | 0.32  | 0.15               |
| (Commitments – retail commitments)/<br>(commitments – retail commitments + loans) | 0.24  | 0.13               |
| Transactions deposits/total deposits                                              | 0.26  | 0.11               |
| <i>Controls for market conditions</i>                                             |       |                    |
| Volatility of S&P 500<br>(annualized return standard deviation)                   | 0.16  | 0.06               |
| Commercial paper–treasury bill yield spread (% pts)                               | 0.40  | 0.21               |
| Yield on 3-month treasury bill (% pts)                                            | 4.52  | 1.55               |
| <i>Controls for bank characteristics</i>                                          |       |                    |
| Assets (billions of \$)                                                           | 34.47 | 69.89              |
| (Cash + securities)/assets                                                        | 0.32  | 0.12               |
| Fed funds purchased/assets                                                        | 0.09  | 0.06               |
| Equity/assets                                                                     | 0.08  | 0.04               |

Summary statistics for all of the variables used in regressions next. Stock-return volatility is equal to the annualized return standard deviation. Data on unused commitments and transactions deposits, as well as the other bank characteristics, come from the most recent quarter of the Reports of Income and Condition prior to the time at which we measure the stock-return variability.

*Table 2: summary statistics*

## 6.3 Tables 3 and 4: main regressions and realized-volatility robustness

**Read:**

- Table 3 shows the benchmark GARCH volatility regressions.
- Commitment exposure has a positive coefficient, while the interaction between commitments and transaction deposits is negative and highly significant.
- Table 4 repeats the analysis using realized volatility and obtains very similar interaction effects.

**Economic message:** The main hedge does not depend on the GARCH volatility model. Deposits reduce the risk effect of commitments under alternative risk measures.

**Table 3**  
Regressions of bank stock-return volatility on liquidity exposure and transactions deposits ratio

|                                                                                        | Dependent variable:<br>Log of weekly bank stock-return volatility<br>predicted volatility from GARCH (1,1) |                    |                    |                   |
|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|--------------------|--------------------|-------------------|
|                                                                                        | (1)                                                                                                        | (2)                | (3)                | (4)               |
| <i>Unused commitments and deposits</i>                                                 |                                                                                                            |                    |                    |                   |
| Commitments/(commitments + loans)                                                      | 0.788<br>(7.19)**                                                                                          | 0.82<br>(6.94)**   | -                  | -                 |
| (Commitments – retail commitments)/<br>(commitments – retail commitments + loans)      | -                                                                                                          | -                  | 0.951<br>(3.55)**  | 1.005<br>(3.63)** |
| Transactions deposits/total deposits                                                   | 0.701<br>(4.66)**                                                                                          | 0.772<br>(3.37)**  | 0.485<br>(1.90)    | 0.497<br>(1.80)   |
| Commitments/(commitments + loans)*                                                     | -1.604<br>(4.46)**                                                                                         | -1.749<br>(3.52)** | -                  | -                 |
| Transactions deposits/total deposits<br>(commitments – RC)/(commitments – RC + loans)* | -                                                                                                          | -                  | -1.702<br>(2.19)*  | -1.709<br>(2.07)* |
| <i>Controls for market conditions</i>                                                  |                                                                                                            |                    |                    |                   |
| Log of volatility of S&P 500                                                           | 0.494<br>(21.43)**                                                                                         | 0.499<br>(20.82)** | 0.464<br>(16.80)** | 0.46<br>(16.43)** |
| Paper-bill spread                                                                      | 0.085<br>(3.79)**                                                                                          | 0.084<br>(3.80)**  | 0.103<br>(4.53)**  | 0.103<br>(4.51)** |
| Yield on 3-month treasury bill                                                         | 0.031<br>(6.96)**                                                                                          | 0.03<br>(6.44)**   | 0.031<br>(6.76)**  | 0.03<br>(6.44)**  |
| <i>Controls for bank characteristics</i>                                               |                                                                                                            |                    |                    |                   |
| Log of assets                                                                          | -0.001<br>(0.06)                                                                                           | -0.003<br>(0.23)   | -0.008<br>(0.57)   | -0.012<br>(0.85)  |
| (Cash + securities)/assets                                                             | -                                                                                                          | -0.03<br>(0.25)    | -                  | -0.108<br>(0.83)  |
| Fed funds purchased/assets                                                             | -                                                                                                          | 0.067<br>(0.26)    | -                  | -0.098<br>(0.37)  |
| Equity/assets                                                                          | -                                                                                                          | -0.174<br>(0.41)   | -                  | 0.095<br>(0.22)   |
| Observations                                                                           | 49,527                                                                                                     | 49,527             | 49,527             | 49,527            |
| Number of independent clusters (banks)                                                 | 170                                                                                                        | 170                | 170                | 170               |
| R <sup>2</sup>                                                                         | 28.77%                                                                                                     | 28.80%             | 27.86%             | 28.02%            |

Regressions from Equation (2) using all data. The dependent variable equals the GARCH (1,1) estimate of conditional stock-return volatility. There are four specifications, two based on the total commitments ratio, and two that exclude retail commitments. One of each specification is a simple model with just log of assets as a bank control, and the other is a more complex model that adds the liquid assets ratio (cash + securities), the ratio of Fed funds purchased to assets, and the capital-asset ratio. Parentheses report absolute value of t-statistics, which are based on robust standard errors clustered by bank, with \* denoting significance at 5%; \*\*

*Table 3: benchmark volatility regressions*

**Table 4**  
Regressions of bank stock-return volatility on liquidity exposure and transactions deposits ratio

|                                                                                        | Dependent variable:<br>Log of weekly bank stock-return volatility<br>Realized volatility |                    |                    |                    |
|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------------------|--------------------|--------------------|
|                                                                                        | (1)                                                                                      | (2)                | (3)                | (4)                |
| <i>Unused commitments and deposits</i>                                                 |                                                                                          |                    |                    |                    |
| Commitments/(commitments + loans)                                                      | 0.803<br>(7.77)**                                                                        | 0.859<br>(7.05)**  | -                  | -                  |
| (Commitments – retail commitments)/<br>(commitments – retail commitments + loans)      | -                                                                                        | -                  | 0.904<br>(3.33)**  | 0.991<br>(3.31)**  |
| Transactions deposits/total deposits                                                   | 0.754<br>(6.04)**                                                                        | 0.899<br>(3.53)**  | 0.545<br>(2.37)*   | 0.621<br>(2.03)*   |
| Commitments/(commitments + loans)*                                                     | -1.541<br>(5.00)**                                                                       | -1.811<br>(3.36)** | -                  | -                  |
| Transactions deposits/total deposits<br>(commitments – RC)/(commitments – RC + loans)* | -                                                                                        | -                  | -1.633<br>(2.20)*  | -1.782<br>(1.98)*  |
| <i>Controls for Market Conditions</i>                                                  |                                                                                          |                    |                    |                    |
| Log of volatility of S&P 500                                                           | 0.511<br>(22.01)**                                                                       | 0.522<br>(20.62)** | 0.481<br>(17.35)** | 0.483<br>(15.84)** |
| Paper-bill spread                                                                      | 0.234<br>(8.59)**                                                                        | 0.231<br>(8.54)**  | 0.253<br>(9.50)**  | 0.25<br>(9.38)**   |
| Yield on 3-month treasury bill                                                         | 0.014<br>(3.17)**                                                                        | 0.013<br>(2.61)**  | 0.014<br>(2.98)**  | 0.013<br>(2.64)**  |
| <i>Controls for bank characteristics</i>                                               |                                                                                          |                    |                    |                    |
| Log of assets                                                                          | 0.001<br>(0.03)                                                                          | -0.003<br>(0.24)   | -0.005<br>(0.32)   | -0.009<br>(0.60)   |
| (Cash + securities)/assets                                                             | -                                                                                        | -0.039<br>(0.30)   | -                  | -0.1<br>(0.70)     |
| Fed funds purchased/assets                                                             | -                                                                                        | -0.016<br>(0.06)   | -                  | -0.187<br>(0.66)   |
| Equity/assets                                                                          | -                                                                                        | -0.379<br>(0.78)   | -                  | -0.113<br>(0.23)   |
| Observations                                                                           | 49,527                                                                                   | 49,527             | 49,527             | 49,527             |
| Number of independent clusters (banks)                                                 | 170                                                                                      | 170                | 170                | 170                |
| R <sup>2</sup>                                                                         | 12.37%                                                                                   | 12.41%             | 11.80%             | 11.87%             |

Regressions from Equation (2) using all data. The dependent variable equals the log of average-squared daily returns over 1 week. There are four specifications, two based on the total commitments ratio, and two that exclude retail commitments. One of each two specifications is a simple model with just log of assets as a bank control, and the other is a more complex model that adds the liquid assets ratio (cash + securities), the ratio of Fed funds purchased to assets, and the capital-asset ratio. Parentheses report absolute value of t-statistics, which are based on robust standard errors clustered by bank, with \* denoting significance at 5%; \*\* significance at 1%. All

*Table 4: realized-volatility robustness*

## 6.4 Tables 5 and 6: residual volatility and between/within variation

Read:

- Table 5 uses residual volatility after removing market, credit, and interest-rate factors.
- The deposit-commitment interaction remains negative and significant.
- Table 6 decomposes the evidence into between-bank and within-bank variation.
- Both dimensions support the hedging effect, though the magnitude differs across specifications.

**Economic message:** The liquidity hedge is not a simple market-beta, credit-risk, or cross-sectional omitted-variable story.

**Table 5**  
Regressions of bank stock-return volatility on liquidity exposure and transactions deposits ratio

|                                                                                     | Dependent variable:<br>Log of weekly bank stock-return volatility<br>realized residual volatility |                    |                    |                    |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|--------------------|--------------------|--------------------|
|                                                                                     | (1)                                                                                               | (2)                | (3)                | (4)                |
| <i>Unused commitments and deposits</i>                                              |                                                                                                   |                    |                    |                    |
| Commitments/(commitments + loans)                                                   | 0.641<br>(6.55)**                                                                                 | 0.771<br>(6.84)**  | -                  | -                  |
| (Commitments – retail commitments)/<br>(commitments – retail commitments + loans)   | -                                                                                                 | -                  | 0.829<br>(3.27)**  | 1.012<br>(3.59)**  |
| Transactions deposits/total deposits                                                | 0.655<br>(5.88)**                                                                                 | 0.992<br>(5.08)**  | 0.508<br>(3.33)**  | 0.766<br>(3.44)**  |
| Commitments/(commitments + loans) *                                                 | -1.217<br>(4.29)**                                                                                | -1.897<br>(4.53)** | -                  | -                  |
| Transactions deposits/total deposits<br>(Commitments-RC)/(commitments-RC + loans) * | -                                                                                                 | -                  | -1.424<br>(2.20)*  | -2.033<br>(1.98)** |
| <i>Controls for market conditions</i>                                               |                                                                                                   |                    |                    |                    |
| Log of volatility of S&P 500                                                        | 0.368<br>(16.25)**                                                                                | 0.392<br>(16.53)** | 0.341<br>(13.52)** | 0.359<br>(13.52)** |
| Paper-bill spread                                                                   | 0.231<br>(9.33)**                                                                                 | 0.227<br>(9.13)**  | 0.247<br>(10.22)** | 0.242<br>(10.00)** |
| Yield on 3-month treasury bill                                                      | 0.04<br>(8.33)**                                                                                  | 0.036<br>(7.00)**  | 0.04<br>(8.14)**   | 0.036<br>(6.93)**  |
| <i>Controls for bank characteristics</i>                                            |                                                                                                   |                    |                    |                    |
| Log of assets                                                                       | -0.041<br>(3.71)**                                                                                | -0.046<br>(3.79)** | -0.048<br>(3.92)** | -0.057<br>(4.09)** |
| (Cash + securities)/assets                                                          | -                                                                                                 | -0.042<br>(0.31)   | -                  | -0.127<br>(0.90)   |
| Fed funds purchased/assets                                                          | -                                                                                                 | 0.086<br>(0.32)    | -                  | -0.062<br>(0.23)   |
| Equity/assets                                                                       | -                                                                                                 | -0.907<br>(2.38)*  | -                  | -0.682<br>(1.62)*  |
| Observations                                                                        | 49,527                                                                                            | 49,527             | 49,527             | 49,527             |
| Number of independent clusters (banks)                                              | 170                                                                                               | 170                | 170                | 170                |
| R <sup>2</sup>                                                                      | 9.05%                                                                                             | 9.27%              | 8.93%              | 9.09%              |

Regressions from Equation (2) using all data. The dependent variable equals the log of the average-squared residual return from the factor model from Equation (3), removing market, credit, and interest rate factors. There are four specifications, two based on the total commitments ratio, and two that exclude retail commitments. One of each specification is a simple model with just log of assets as a bank control, and the other is a more complex model that adds the liquid assets ratio (cash + securities), the ratio of Fed funds purchased to assets, and the capital-asset ratio. Parentheses report absolute value of t-statistics, which are based on robust standard

*Table 5: residual-volatility regressions*

**Table 6**  
Regressions of bank stock-return volatility on liquidity exposure and transactions deposits ratio—between versus within estimator

|                                          | Dependent variable:<br>Log of weekly bank stock-return volatility |                    |                    |                    |
|------------------------------------------|-------------------------------------------------------------------|--------------------|--------------------|--------------------|
|                                          | Between estimator                                                 |                    | Within estimator   |                    |
|                                          | (1)                                                               | (2)                | (3)                | (4)                |
| <i>Unused commitments and deposits</i>   |                                                                   |                    |                    |                    |
| Commitments/(commitments + loans)        | 0.762<br>(4.58)**                                                 | 0.805<br>(3.94)**  | 0.202<br>(1.41)    | 0.082<br>(0.60)    |
| Transactions deposits/total deposits     | 0.815<br>(3.55)**                                                 | 1.013<br>(2.98)**  | 0.281<br>(0.98)    | 0.237<br>(0.85)    |
| Commitments/(commitments + loans) *      | -1.384<br>(2.00)*                                                 | -1.739<br>(2.08)*  | -1.668<br>(2.64)** | -1.492<br>(2.53)*  |
| <i>Controls for market conditions</i>    |                                                                   |                    |                    |                    |
| Log of volatility of S&P 500             | 0.549<br>(6.52)**                                                 | 0.569<br>(6.47)**  | 0.447<br>(24.61)** | 0.452<br>(25.01)** |
| Paper-bill spread                        | -1.84<br>(4.48)**                                                 | -1.783<br>(4.22)** | 0.163<br>(8.63)**  | 0.162<br>(8.60)**  |
| Yield on 3-month treasury bill           | 0.204<br>(5.39)**                                                 | 0.195<br>(4.85)**  | 0.024<br>(5.11)**  | 0.024<br>(5.15)**  |
| <i>Controls for bank characteristics</i> |                                                                   |                    |                    |                    |
| Log of assets                            | -0.014<br>(0.83)                                                  | -0.014<br>(0.72)   | -0.014<br>(0.59)   | 0.011<br>(0.41)    |
| (Cash + securities)/assets               | -                                                                 | 0.057<br>(0.38)    | -                  | 0.225<br>(1.59)    |
| Fed funds purchased/assets               | -                                                                 | -0.013<br>(0.04)   | -                  | -0.458<br>(1.93)   |
| Equity/assets                            | -                                                                 | -0.527<br>(0.84)   | -                  | -1.125<br>(2.22)*  |
| Observations                             | 49,527                                                            | 49,527             | 49,527             | 49,527             |
| Number of independent clusters (banks)   | 170                                                               | 170                | 170                | 170                |
| R <sup>2</sup>                           | 39.81%                                                            | 40.15%             | 29.73%             | 30.28%             |

Regressions from Equation (2) using all data. There are four specifications: columns 1 and 2 report the “between” estimator from regressions using bank-level time-series averages, and columns 3 and 4 report the “within” estimator, which allows each bank to have its own intercept in the regressions. One of each two specifications is a simple model with just log of assets as a bank control, and the other is a more complex model that adds the liquid assets ratio (cash + securities), the ratio of Fed funds purchased to assets, and the capital-asset ratio. Parentheses report absolute value of t-statistics, which are based on robust standard errors clustered by bank, with \*denoting significance at 5%, \*\*significance at 1%. All regressions include an intercept

*Table 6: between versus within estimator*

## 6.5 Table 7 and Figure 3: additional controls and 1998 crisis scatter

Read:

- Table 7 adds credit-risk, market-risk, and time controls.
- The key interaction between commitments and deposits remains negative and statistically significant.
- Figure 3 focuses on fall 1998. For low-deposit banks, the commitment-volatility slope increases to 0.46; for high-deposit banks, the slope remains negative and insignificant.

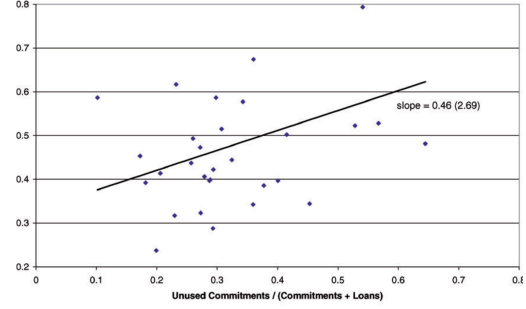
**Economic message:** The hedge becomes more valuable in crisis, exactly when borrowers draw credit lines and transactions deposits flow into banks.

**Table 7**  
Regressions of bank stock-return volatility on liquidity exposure and transactions deposits ratio—add controls for credit risk, market risk, and time dummies

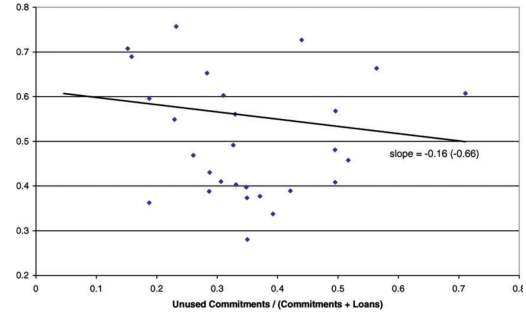
|                                             | Dependent variable:<br>Log of weekly bank stock-return volatility |                    |                    |                    |
|---------------------------------------------|-------------------------------------------------------------------|--------------------|--------------------|--------------------|
|                                             | (1)                                                               | (2)                | (3)                | (4)                |
| <i>Unused commitments and deposits</i>      |                                                                   |                    |                    |                    |
| Commitments/(commitments + loans)           | 0.610<br>(5.28)**                                                 | 0.507<br>(4.31)**  | 0.600<br>(5.76)**  | 0.501<br>(4.67)**  |
| Transactions deposits/total deposits        | 0.279<br>(1.55)                                                   | 0.251<br>(1.51)    | 0.427<br>(2.53)**  | 0.418<br>(2.61)**  |
| Commitments/(commitments + loans) *         | -1.067<br>(2.45)*                                                 | -0.884<br>(2.21)*  | -0.921<br>(2.38)*  | -0.800<br>(2.18)*  |
| <i>Controls for market conditions</i>       |                                                                   |                    |                    |                    |
| Log of volatility of S&P 500                | 0.507<br>(22.95)**                                                | 0.498<br>(22.32)** | -                  | -                  |
| Paper-bill spread                           | 0.100<br>(4.35)**                                                 | 0.111<br>(4.76)**  | -                  | -                  |
| Yield on 3-month treasury bill              | 0.029<br>(5.67)**                                                 | 0.027<br>(5.63)**  | -                  | -                  |
| <i>Controls for bank characteristics</i>    |                                                                   |                    |                    |                    |
| Log of assets                               | -0.012<br>(0.96)                                                  | -0.007<br>(0.64)   | -0.015<br>(1.21)   | -0.012<br>(1.04)   |
| (Cash + securities)/assets                  | 0.246<br>(1.73)                                                   | 0.332<br>(2.36)*   | 0.125<br>(0.96)    | 0.213<br>(1.68)    |
| Fed funds purchased/assets                  | 0.141<br>(0.58)                                                   | 0.144<br>(0.60)    | 0.163<br>(0.72)    | 0.166<br>(0.74)    |
| Equity/assets                               | 0.584<br>(2.37)*                                                  | 0.689<br>(3.04)**  | 0.552<br>(2.30)*   | 0.594<br>(2.46)*   |
| <i>Controls for credit and market risks</i> |                                                                   |                    |                    |                    |
| C&I loans/assets                            | 0.369<br>(2.29)*                                                  | 0.417<br>(2.66)**  | 0.230<br>(1.33)    | 0.287<br>(1.71)    |
| Commercial real estate loans/assets         | -0.337<br>(1.31)                                                  | -0.179<br>(0.71)   | -0.454<br>(1.70)   | -0.321<br>(1.24)   |
| Nonperforming loans/assets                  | 10.832<br>(5.66)**                                                | 7.859<br>(4.37)**  | 8.890<br>(4.33)**  | 6.476<br>(3.63)**  |
| Trading assets/assets                       | 0.345<br>(2.00)*                                                  | 0.455<br>(2.79)**  | 0.320<br>(2.08)*   | 0.415<br>(2.81)**  |
| AAA-rated indicator                         | -0.142<br>(2.58)**                                                | -0.147<br>(2.78)** | -0.259<br>(4.42)** | -0.250<br>(4.44)** |
| AA-rated indicator                          | -0.075<br>(1.74)                                                  | -0.075<br>(1.78)   | -0.103<br>(2.29)*  | -0.098<br>(2.27)*  |
| A-rated indicator                           | -0.095<br>(3.70)**                                                | -0.095<br>(3.85)** | -0.096<br>(3.72)** | -0.094<br>(3.80)** |
| Unrated indicator                           | -0.032<br>(1.05)                                                  | -0.031<br>(1.05)   | -0.047<br>(1.56)   | -0.043<br>(1.48)   |
| Loan loss provisions/assets                 | -                                                                 | 22.923<br>(5.14)** | -                  | 20.173<br>(4.92)** |
| Net charge-offs/assets                      | -                                                                 | 14.411<br>(1.98)*  | -                  | 15.645<br>(2.30)*  |
| Includes weekly indicators?                 | No                                                                | No                 | Yes                | Yes                |
| Observations                                | 49,527                                                            | 49,527             | 49,527             | 49,527             |
| Number of independent clusters (banks)      | 170                                                               | 170                | 170                | 170                |
| R <sup>2</sup>                              | 37.47%                                                            | 38.52%             | 50.46%             | 51.31%             |

Regressions from Equation (2) using all data and including control variables for market conditions, bank characteristics, and credit and market risks. There are four specifications: those in columns 1 and 2 use time-varying control variables (S&P 500 volatility, the level of interest rates, and the paper-bill

*Table 7: credit and market controls*



**Figure 3a**  
Stock Return Volatility for Low Transactions Deposit Banks, Fall of 1998.



**Figure 3b**  
Stock Return Volatility for High Transactions Deposit Banks, Fall of 1998. Source: Volatility based on authors' calculations using data from CRSP. Commitment ratio is from Call Reports.

commitments comes with a 7% point increase (14% of mean volatility during

*Figure 3: fall 1998 crisis scatter*

## 6.6 Tables 8 and 9: high-spread regimes and continuous spread interactions

Read:

- Table 8 splits the sample into low and high paper-bill spread regimes.
- The interaction term becomes much more negative when spreads are high and is largest during October 1998.
- Table 9 interacts commitments and transaction deposits with the paper-bill and credit spreads directly.
- The triple-interaction evidence confirms that the hedge strengthens when market liquidity is tight.

**Economic message:** Market conditions are central. Deposit-loan synergies are not constant; they become stronger when liquidity insurance is most valuable.

**Table 8**  
Regressions of bank stock-return volatility on liquidity exposure and transactions deposits ratio—high and low paper-bill spread regimes

|                                                    | Dependent variable:<br>Log of weekly bank stock-return volatility |                    |                                    |                            |
|----------------------------------------------------|-------------------------------------------------------------------|--------------------|------------------------------------|----------------------------|
|                                                    | <75 Bps<br>(1)                                                    | ≥75 Bps<br>(2)     | P value<br>$H_0: (1) = (2)$<br>(3) | Data from<br>Oct-98<br>(4) |
| Paper-bill spread:                                 |                                                                   |                    |                                    |                            |
| <i>Unused commitments and deposits</i>             |                                                                   |                    |                                    |                            |
| Commitments/(commitments + loans)                  | 0.761<br>(8.12)**                                                 | 1.002<br>(4.67)**  | 0.212                              | 1.893<br>(5.93)**          |
| Transactions deposits/total deposits               | 0.632<br>(3.35)**                                                 | 1.396<br>(3.11)**  | 0.057                              | 3.743<br>(3.29)**          |
| Commitments/(commitments + loans) *                | -1.317<br>(3.42)**                                                | -3.196<br>(3.47)** | 0.02                               | -7.453<br>(3.95)**         |
| <i>Controls for bank characteristics</i>           |                                                                   |                    |                                    |                            |
| Log of assets                                      | -0.006<br>(0.56)                                                  | 0.042<br>(2.53)*   | 0.001                              | 0.039<br>(0.91)            |
| (Cash + securities)/assets                         | -0.063<br>(0.60)                                                  | -0.304<br>(1.74)   | 0.12                               | -0.523<br>(1.24)           |
| Fed funds purchased/assets                         | 0.129<br>(0.52)                                                   | 0.316<br>(0.88)    | 0.617                              | -0.494<br>(0.64)           |
| Equity/assets                                      | 0.411<br>(1.38)                                                   | -0.765<br>(1.18)   | 0.043                              | -3.507<br>(2.66)**         |
| P-value for F-test that all coefficients are equal | -                                                                 | -                  | 0.001                              | -                          |
| Includes weekly indicators?                        | Yes                                                               | Yes                | -                                  | Yes                        |
| Observations                                       | 46,856                                                            | 2,671              | -                                  | 192                        |
| Number of independent clusters (banks)             | 170                                                               | 154                | -                                  | 64                         |
| R <sup>2</sup>                                     | 43.53%                                                            | 39.39%             | -                                  | 30.30%                     |

Regressions in two regimes: one when the paper-bill spread is below 75 basis points (column 1) and the other when the spread is above 75 basis points (column 2). Column 3 reports the *p*-value for the test that the respective coefficients are equal in the two regimes. Column 4 includes data during the first 3 weeks of October 1998, when the paper-bill spread exceeded 100 basis points. Parentheses report absolute value of

Table 8: high and low paper-bill spread regimes

## 7 Short summary

- Banks provide liquidity both to depositors and to borrowers with loan commitments.
- Unused commitments raise bank stock-return volatility, reflecting asset-side liquidity risk.
- Transaction deposits reduce the risk effect of unused commitments.
- The deposit-commitment interaction is negative and significant across GARCH, realized-volatility, and residual-volatility specifications.
- The result survives controls for size, liquid assets, capital, fed-funds borrowing, credit risk, market risk, and time effects.
- The hedge operates in both between-bank and within-bank variation.
- The hedge becomes stronger when commercial-paper spreads are high and during the 1998 crisis.
- The paper shows that transactions deposits can stabilize banks, reversing the usual view that deposits only create liquidity risk.

## 8 Conclusion

### 8.1 Core conclusion

Gatev, Schuermann, and Strahan show that deposits and loan commitments are jointly valuable because they hedge each other. Loan commitments expose banks to borrower liquidity demand, while transactions deposits provide liquidity inflows, especially in market stress. Banks with strong transactions deposit bases can therefore offer loan commitments with less equity-market risk.

### 8.2 Economic intuition

The key intuition is risk pooling across the two sides of the bank balance sheet. Borrowers draw credit lines when outside funding markets are tight. At the same time, investors and households often move funds into banks. A bank that combines both activities benefits from naturally offsetting liquidity flows.

**Table 9**  
Regressions of bank stock-return volatility on liquidity exposure and transactions deposits ratio—with paper-bill spread interaction terms

|                                                  | Dependent variable:<br>Log of weekly bank stock-return volatility<br>Realized residual volatility |                    |                    |                   |
|--------------------------------------------------|---------------------------------------------------------------------------------------------------|--------------------|--------------------|-------------------|
|                                                  | (1)                                                                                               | (2)                | (3)                | (4)               |
| <i>Unused commitments and deposits</i>           |                                                                                                   |                    |                    |                   |
| Commitments/(commitments + loans)                | 0.558<br>(2.80)**                                                                                 | 0.542<br>(2.85)**  | 0.790<br>(4.59)**  | 0.735<br>(3.13)** |
| Transactions deposits/total deposits             | 0.364<br>(2.00)*                                                                                  | 0.421<br>(2.15)*   | 0.564<br>(1.69)    | 0.523<br>(0.95)   |
| Commitments/(commitments + loans) *              | -0.370<br>(0.70)                                                                                  | -0.467<br>(0.85)   | -0.209<br>(0.29)   | -0.330<br>(0.31)  |
| <i>Interaction between paper-bill spread and</i> |                                                                                                   |                    |                    |                   |
| Commitments/(commitments + loans)                | 0.323<br>(0.88)                                                                                   | 0.591<br>(1.49)    | 0.183<br>(0.59)    | 0.436<br>(1.16)   |
| Transactions deposits/total deposits             | 1.101<br>(3.05)**                                                                                 | 1.508<br>(2.87)**  | 0.908<br>(2.56)**  | 1.284<br>(2.22)*  |
| Commitments/(commitments + loans) *              | -2.825<br>(2.84)**                                                                                | -3.538<br>(2.84)** | -2.557<br>(2.71)** | -3.137<br>(2.36)* |
| <i>Interaction between Baa-Aaa spread and:</i>   |                                                                                                   |                    |                    |                   |
| Commitments/(commitments + loans)                | -                                                                                                 | -                  | -0.210<br>(1.11)   | -0.160<br>(0.76)  |
| Transactions deposits/total deposits             | -                                                                                                 | -                  | -0.149<br>(0.45)   | -0.056<br>(0.12)  |
| Commitments/(commitments + loans) *              | -                                                                                                 | -                  | -0.340<br>(0.46)   | -0.320<br>(0.34)  |
| Controls for bank characteristics                | <i>Included, coefficients not reported</i>                                                        |                    |                    |                   |
| Includes weekly indicators?                      | Yes                                                                                               | Yes                | Yes                | Yes               |
| Observations                                     | 49,527                                                                                            | 49,527             | 48,991             | 48,991            |
| Number of Independent clusters (banks)           | 170                                                                                               | 170                | 170                | 170               |
| R <sup>2</sup>                                   | 24.92%                                                                                            | 25.06%             | 25.05%             | 25.25%            |

In regressions using all data, we added interactions between the level of the paper-bill spread and the transaction deposits and loan commitment variables (columns 1 and 2) and interaction

Table 9: spread interaction regressions

### **8.3 Takeaway**

The paper provides direct evidence for the liquidity-production synergy at the heart of bank specialness. Banks are not special merely because they make loans or issue deposits separately. They are special because combining the two products reduces liquidity risk, especially in crises.